

Customer Churn Analysis Project

In this project, I performed a comprehensive exploratory data analysis (EDA) on a telecom dataset to understand **customer churn behavior**—specifically, why some customers choose to leave the company. This project sharpened my skills in **data preprocessing**, **EDA**, and **data storytelling** using visualizations.

Data Preparation

- Loaded the dataset using **pandas** and explored the structure with `.info()` and `.describe()`.
- Handled missing/invalid values in the **TotalCharges** column by replacing blanks with 0 and converting the column to float.
- Checked for duplicate entries and null values.
- Converted **SeniorCitizen** from numeric to categorical (0 → No, 1 → Yes).

Exploratory Data Analysis (EDA)

Used **seaborn** and **matplotlib** to visualize the relationship between churn and various customer attributes:

Churn Distribution:

- **Bar Chart & Pie Chart** revealed:
 - 26% of customers have churned
 - 74% have stayed

Customer Demographics:

- **Gender:** Similar churn rates for both genders.
- **Senior Citizens:** Higher churn rate compared to non-senior customers.

Services & Subscriptions:

- Customers with **fiber optic internet**, **no tech support**, **no online security**, or **streaming services** showed increased likelihood to churn.

Contract & Billing:

- **Month-to-month contracts** saw the **highest churn**.
- Longer contracts (1 or 2 years) showed better customer retention.
- **Electronic check** as a payment method was common among churned customers.

Numerical Features:

- Churn is higher among customers with:
 - **Higher Monthly Charges**
 - **Lower Tenure**
 - Visualized using **boxplots** and **histograms**
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Key Takeaways:

- **High churn risk** is linked to short tenure, high bills, and lack of support services.
 - Customers with **monthly contracts and fiber internet** are more prone to leave.
 - Visuals effectively highlight churn-prone customer segments.
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